SalesGenie AI — API Contract

Status: Proposed contract for team review
Version: 0.1
Prepared: 15 October 2026
Base path: `/api/v1`
Related specification: SRS.md

> This contract is ready for backend/database review, not yet formally approved. The API is intended for local development; it must not be publicly exposed with real lead data before an authentication/security design is approved.

1. Conventions

- JSON request and response bodies use `application/json`, except CSV upload.
- Identifiers are integer `lead_id` values.
- Timestamps are ISO 8601 UTC strings.
- Unknown fields and invalid values return a validation error; they are not silently ignored.
- List endpoints return stable pagination metadata.
- Errors use the common envelope below.

1.1 Error response

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The request contains invalid fields.",
    "details": [
      {"field": "email", "message": "Must be a valid email address."}
    ]
  }
}
```

`details` may be an empty array when there are no field-specific details. Do not return secrets, provider credentials, stack traces, or internal database details.

2. Lead schema

```
{
  "lead_id": 123,
  "company_name": "Example Systems",
  "industry": "Software",
  "contact_name": "Jordan Lee",
  "email": "jordan@example.com",
  "phone": "+1-555-0100",
  "lead_status": "New",
  "created_at": "2026-10-15T12:00:00Z",
  "updated_at": "2026-10-15T12:00:00Z"
}
```

`company_name` is required. Other contact/company fields are optional and nullable. The server owns `lead_id`, `created_at`, and `updated_at`; clients must not set them.

Proposed `lead_status` values: `New`, `Contacted`, `Qualified`, `Proposal`, `Won`, `Lost`. Backend and database owners must confirm these values before implementation.

3. Milestone 1 endpoints

3.1 Create lead

`POST /api/v1/leads`

Request:

```
{
```

```
  "company_name": "Example Systems",
  "industry": "Software",
  "contact_name": "Jordan Lee",
  "email": "jordan@example.com",
  "phone": "+1-555-0100"
}
```

Response: `201 Created`, body is the created Lead schema.

Errors: `400` validation failure; `409` duplicate detected where the confirmed duplicate rule applies.

3.2 List, search, and filter leads

`GET /api/v1/leads`

Query parameters:

Name	Type	Default	Meaning
`q`	string	none	Case-insensitive search over company, contact, email, and industry
`status`	string	none	Filter by an allowed lifecycle status
`page`	integer ≥ 1	`1`	Page number
`page_size`	integer 1–100	`20`	Maximum records in the page

Response: `200 OK`.

```
{
  "items": [],
  "page": 1,
  "page_size": 20,
  "total": 0
}
```

3.3 Retrieve lead

`GET /api/v1/leads/{lead_id}`

Response: `200 OK`, body is the Lead schema. Unknown ID: `404 Not Found`.

3.4 Update lead

`PATCH /api/v1/leads/{lead_id}`

Request contains one or more editable fields from `company_name`, `industry`, `contact_name`, `email`, and `phone`. `lead_id`, timestamps, and `lead_status` are not editable through this endpoint.

Response: `200 OK`, body is the updated Lead schema. Unknown ID: `404`; invalid fields: `400`.

3.5 Update lifecycle status

`PATCH /api/v1/leads/{lead_id}/status`

Request:

```
{"lead_status": "Qualified"}
```

Response: `200 OK`, body is the updated Lead schema. Unknown ID: `404`; unsupported status: `400`.

3.6 Import CSV

`POST /api/v1/leads/import`

Content type: `multipart/form-data`; field name: `file`. CSV header names: `company_name`, `industry`, `contact_name`, `email`, `phone`, `lead_status`. `company_name` is required; if `lead_status` is omitted, it defaults to `New`.

Response: `200 OK`.

```
{
  "created": 2,
```

```
  "duplicates": 1,
  "rejected": 1,
  "row_errors": [
    {"row": 5, "code": "VALIDATION_ERROR", "message": "company_name is required."}
  ]
}
```

Rows with invalid values must be reported. The proposed duplicate rule is normalized email when present, otherwise normalized company name plus contact name; team confirmation is required. Import processing must not report rejected or duplicate rows as created.

3.7 Analyze a lead's company

`POST /api/v1/leads/{lead_id}/analysis`

Request:

```
{
  "context": "Optional user-supplied company or industry notes."
}
```

`context` is optional. The first milestone uses saved lead fields and supplied context; live web research is not promised by this contract.

Response: `201 Created`.

```
{
  "analysis_id": 456,
  "lead_id": 123,
  "summary": "Concise generated summary.",
  "potential_needs": ["Example need"],
  "opportunities": ["Example opportunity"],
  "qualification": {
    "score": null,
    "rationale": "Insufficient evidence for a numeric qualification."
  },
  "created_at": "2026-10-15T12:05:00Z"
}
```

The analysis must be persisted and linked to the requested lead. `qualification.score` is nullable and, when present, is an integer from 0 to 100. It is an estimate, not a Milestone 2 accuracy claim.

Errors: `404` unknown lead; `422` invalid request; `502` upstream AI provider failure. Provider failures must be explicit and must not create a success-shaped analysis record.

3.8 Retrieve the latest saved analysis

`GET /api/v1/leads/{lead_id}/analysis`

Response: `200 OK`, body is the latest analysis schema. Unknown lead: `404`; no analysis exists: `404` with code `ANALYSIS_NOT_FOUND`.

4. Future module endpoint proposals

These paths reserve a consistent API shape for the six-module scope. They are not part of the Milestone 1 implementation commitment and require detailed contracts before their milestone starts.

Module	Proposed endpoint	Milestone
Outreach generation	`POST /api/v1/leads/{lead_id}/outreach-drafts`	2
Lead score/recommendation	`POST /api/v1/leads/{lead_id}/scores` and `GET /api/v1/leads/{lead_id}/scores/latest`	2
Conversation summary	`POST /api/v1/leads/{lead_id}/conversation-summaries`	3
CRM integration	Vendor-specific integration routes to be agreed	3

Module	Proposed endpoint	Milestone
Dashboard analytics	`GET /api/v1/analytics/summary`	4

5. Proposed persistence mapping

Resource	Table	Important fields
Lead	`leads`	`lead_id`, company/contact fields, `lead_status`, timestamps
Analysis	`company_insights`	`analysis_id`, `lead_id` foreign key, result fields, `created_at`
Score	`lead_scores`	`score_id`, `lead_id` foreign key, score, rationale, timestamp
Outreach draft	`outreach_drafts`	draft ID, `lead_id` foreign key, type, content, timestamp
Activity	`activities`	activity ID, `lead_id` foreign key, type, details, timestamp
Conversation summary	`conversation_summaries`	summary ID, optional `lead_id` foreign key, summary, key points, actions, timestamp

Use database migrations for schema changes. The precise JSON/list storage representation must be tested against both the selected SQLite development configuration and PostgreSQL production configuration.

6. Shared decisions to close before coding

1. Approve lifecycle values and duplicate matching rule.
2. Confirm CSV encoding, maximum upload size, and whether imports are all-or-partial on row errors.
3. Confirm whether a numeric qualification score is in the Milestone 1 API response.
4. Confirm OpenAI model and provider timeout/retry policy.
5. Agree on lead/analysis response types with the Streamlit consumer.
6. Record a versioned OpenAPI schema alongside implementation.